

BIG DATA

[CSE] Big Data

[DTM] Big Data and Cloud Platforms

About us



[Matteo Francia](#), Ph.D.

- Tenure-Track Researcher
- Teacher for
 - DTM (Module 2)
[main teacher]



[Enrico Gallinucci](#), Ph.D.

- Tenure-Track Researcher
- Teacher for
 - CSE (Modules 1 & 2)
[main (and only) teacher]
 - DTM (Module 1)

Members of the [Business Intelligence Group](#)



**CONTACT US FOR
THESIS OPPORTUNITIES**

Program

→ [CSE+DTM Module 1] Theoretical background on Big Data (20 hours)

- HW infrastructures, SW architectures
- Data structures (Parquet) and storage services (DFS, NoSQL)
- Batch applications: MapReduce programming paradigm, Apache Spark, SQL-on-Hadoop
- Streaming algorithms and applications

→ [CSE Module 2] In-depth practice + Big Data in the Cloud (30 hours)

- Main focus: hands-on Apache Spark applications and non-trivial Map Reduce programming
- Industrial seminars

[DTM Module 2] Handling Big Data in the Cloud + guided practice (30 hours)

- Main focus: cloud platforms, including taxonomies, billing models, on-prem vs cloud
- Industrial seminars

Technologies

Local environment + Virtual remote lab

- Hosted by AWS Academy
- 2x \$50 credit (renewable, no credit card required)



Main applications

- Apache Hadoop
- Apache Spark
 - Core
 - SQL
 - Streaming
- Apache Kafka



Programming language

- [CSE] Scala & Python
- [DTM] Python



Do it now!

Subscribe to the course on Virtuale

- Any important information (e.g., changes in the schedule, thesis opportunities) will be sent through Virtuale
- **Invitations to AWS Academy will be sent on October 1 only to who is subscribed**
- If you subscribe later, send me an email to request the invitation to AWS

Schedule

Warning

- Schedule can change
- In such case, communications will be sent

No classes on

- Week 40 (Sept. 28 to Oct. 2)

Office hours

- By appointment

Week #	Monday Room 2.7 1 pm (2h)	Tuesday Lab 3.3 11 am (2h)	Wednesday Lab 4.2 9 am (3h)
38	Module 1	Module 1	Module 1
39	Module 1	Module 1	Module 1
40	---	---	---
41	Module 1	Module 1	Mod. 2 CSE
42	Module 1	Mod. 2 DTM	Mod. 2 CSE
43	Mod. 2 DTM	Mod. 2 DTM	Mod. 2 CSE
44	Mod. 2 DTM	Mod. 2 DTM	Mod. 2 CSE
45	Mod. 2 DTM	Mod. 2 DTM	Mod. 2 CSE
46	Mod. 2 DTM	Mod. 2 DTM	Mod. 2 CSE
47	Mod. 2 DTM	Mod. 2 DTM	Mod. 2 CSE
48	Mod. 2 DTM	Mod. 2 DTM	Mod. 2 CSE
49	---	---	Mod. 2 CSE
50	---	---	Mod. 2 CSE
51	---	---	---

Exam DTM: oral examination

Questions on all (theoretical and practical) aspects of the course

- A single session (~30 min.) with both teachers, each covering his own module
- Seminars are included
- Interaction during the lectures/labs is considered in the final evaluation

According to the University's regulation

- Exams must be in presence
- Cannot refuse a grade more than once

No scheduled dates, just come when you are ready

- Booking is self-service (with a minimum of 1 week notice): [LINK](#)
- Do not book if you are not sure about being prepared
- If you have troubles finding a slot and/or have special needs, write us an email

Be prepared: if you fail or refuse, you will have to wait 1 month to try again

Exam CSE: project + oral examination

Project

- Mandatory
 - **Must be pre-arranged with the teacher**
 - Groups allowed, up to 2 people
 - Guidelines to be published on GitHub
- $\text{vote}(\text{project}) \in [0,2]$
 - Once discussed, the project is final

Oral

- Discussion of project (~15 min.) + questions on all (theoretical and practical) aspects of the course (~30 min.) on the same day
 - Booking is self-service (with a minimum of 1 week notice): [LINK](#)
 - **Do not book if you are not sure about being prepared**
 - If you have troubles finding a slot and/or have special needs, write me an email
- $\text{vote}(\text{oral}) \in [0,30]$

Final vote: $\text{vote}(\text{project}) + \text{vote}(\text{oral})$

Exam CSE: project details

Find a suitable dataset

- One or more tables/collections of data
- Overall, in the order of GBs

Write a Spark program to run an analytical pipeline

- Focus on proper design and performance tuning
- Verify and discuss results

Evaluation

- 0 points for sufficient projects
- 1-2 additional points based on technical complexity, performance evaluation, and creativity

References

Hadoop and Spark

- **Learning Spark. Lightning-Fast Data Analytics (2nd edition)**
Jules S. Damji, Brooke Wenig, Tathagata Das, and Denny Lee; O'Reilly, 2020
- **Hadoop. The Definitive Guide (4th edition)**
Tom White; O'Reilly, 2015
- **Spark: The Definitive Guide. Big Data Processing Made Simple**
Bill Chambers and Matei Zaharia; O'Reilly, 2018

Algorithms and architectures

- **Designing Data-Intensive Applications. The big ideas behind reliable, scalable, and maintainable systems**
Martin Kleppmann; O'Reilly, 2017
- **Architecting Modern Data Platforms. A guide to enterprise Hadoop at scale**
Jan Kunigk, Ian Buss, Paul Wilkinson, and Lars George; O'Reilly, 2019
- **Mining of Massive Datasets (2nd edition)**
Jure Leskovec, Anand Rajaraman, and Jeffrey D. Ullman; Cambridge University Press, 2014

References

Streaming

- **Streaming Data - Understanding the real-time pipeline**
Andrew G. Psaltis; Manning, 2017
- **Streaming Systems. The What, Where, When, and How of Large-Scale Data Processing**
Tyler Akidau, Slava Chernyak, and Reuven Lax; O'Reilly, 2018

Cloud

- **Cloud Computing for Science and Engineering**
Ian Foster, Dennis Gannon; MIT Press, 2017
- **Designing Cloud Data Platforms**
Zburivsky, Danil, Lynda Partner; Simon and Schuster, 2021

NoSQL

- **Database Internals. A Deep Dive into How Distributed Data Systems Work**
Alex Petrov; O'Reilly, 2019
- **Next Generation Databases. NoSQL, NewSQL, and Big Data**
Guy Harrison; Apress, 2015